

AMARTYA KUMAR MAULIK

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OBJECTIVE

Doctoral candidate in Statistics with expertise in Bayesian modeling, MCMC, mixture models, predictive analytics, and biomedical data, seeking a data scientist position focused on advanced modeling and data-driven decision-making.

EDUCATION

Doctor of Philosophy in Statistics, *Colorado State University* August 2022 - May 2027 (Expected)
GPA: 4/4 | Advisor: Dr. Tianjian Zhou

Master of Science in Statistics, *Indian Institute of Technology, Kanpur* August 2019 - May 2021
CGPA: 9.2/10

Bachelor of Science (Hons.) in Statistics, *University of Calcutta* August 2016 - May 2019
CGPA: 8.58/10 | Minor in Mathematics and Computer Science

WORK EXPERIENCE

Graduate Teaching Assistant August 2022 - Present
Colorado State University Fort Collins, CO

- Instructor for courses: Computing and Math Skills for Statistical Analysis; Intro to Applied Statistics.
- Grader for courses: Regression Models and Applications; Mixed Models and Statistical Data Analysis.

Data Analyst Intern, Intelligent Customer Interactions Team May 2020 - July 2020
Ford Motor Company (GDIA) Chennai, India

- Built a customer-level classification model in Python to predict the likelihood of purchasing Ford nameplates.
- Mitigated class imbalance and compared logistic, decision tree, gradient boosting, and random forest models.
- Identified random forest as best-performing (balanced accuracy: 55%).
- Improved segment-level separability by 33% through hyperparameter tuning.

RESEARCH PROJECTS

Hierarchical Sparse Mixture of Finite Mixtures (Ongoing)
Advisor: Dr. Matt Koslovsky and Dr. Tianjian Zhou, Colorado State University May 2025 -

- Developing a hierarchical sparse mixture model with group-specific component activation using zero-inflated priors, including posterior derivations and Gibbs sampling for efficient Bayesian inference.

Bayesian Functional Generalized Probit Mixed Model for Scalar-on-Function Regression (Ongoing)
Advisor: Dr. Tianjian Zhou, Colorado State University May 2025 -

- Developed a Bayesian scalar-on-function regression model for four-category outcomes with spatially indexed functional predictors.
- Incorporated subject- and limb-level random effects to account for within-horse dependence and bilateral variability.
- Applied the model to equine osteochondral disease data to evaluate prediction and estimate spatially varying effects across forelimbs.

A Bayesian Decision-Theoretic Approach to Multiple Testing in Basket Trials [\[Link\]](#)

Published in *Biometrics*
Advisor: Dr. Tianjian Zhou, Colorado State University May 2024 - June 2026

- Designed a scalable Bayesian decision-theoretic framework for multiple hypothesis testing across subpopulations.
- Enabled adaptive information borrowing while retaining independent estimation and low computational cost.
- Calibrated decision rules to satisfy frequentist error control targets relevant to regulatory decision-making.

- Demonstrated performance through extensive simulation studies and a real-world clinical trial case study.

Prediction tool for discharge disposition and 30-day readmission using electronic health records among patients hospitalized for traumatic brain injury [\[Link\]](#)

Published in *Frontiers in Neurology* May 2024 - June 2025
Advisor: Dr. Tianjian Zhou, Colorado State University

- Developed and validated EHR-based models for discharge disposition and 30-day readmission among hospitalized traumatic brain injury patients.
- Analyzed multi-hospital data from 2017-2023 using sociodemographic, clinical, functional, and prior-utilization variables.
- Applied multinomial and logistic regression with AIC-based variable selection and cross-validated AUC evaluation.
- Identified key predictors of discharge and readmission and translated the models into a clinical risk-stratification tool.

Martingales in Discrete Time

Guide: Dr. Supriyo Bhar, Indian Institute of Technology, Kanpur May 2021

- Studied discrete-time martingales using measure-theoretic conditional expectation, including convergence theorems, $\mathcal{L}^2$ -bounded martingales, and uniform integrability.
- Examined the Martingale Representation Lemma and its application to the discrete-time Black–Scholes pricing model.

MCMC in Mixture Models

Guide: Dr. Dootika Vats, Indian Institute of Technology, Kanpur May 2021

- Modeled a simulated statistical distribution by a mixture of other distributions using the infinite Dirichlet mixture model without assuming a finite number of clusters.
- Implemented Collapsed Gibbs, Blocked Gibbs, and Slice Sampling methods for parameter updates in R.

AWARDS

- James R. ZumBrunnen Award (*CSU Department of Statistics*) 2025
Recognizes an outstanding student who has made significant contribution through collaboration with Colorado State University researchers.
- Elmer E. Remmenga Scholarship (*CSU Department of Statistics*) 2024
Intended to encourage students of statistics to develop their interests in applied statistics and statistical consulting.

SKILLS AND INTERESTS

Tools and Languages	R, Python
Modeling	Supervised and Unsupervised Learning (Clustering and Dimension Reduction), Regression, Data Analysis, Bayesian Computation, MCMC
Soft Skills	Communication, Problem Solving, Creativity

ACTIVITIES

- Director of Finance, Indian Students Association Fall 2023 - Spring 2024
- Organizer, Student Organized Activities and Research Seminars (SOARS) Spring 2023 - Fall 2023